

Catalytic Impact of Early Investments in Advanced Clean Energy Technologies

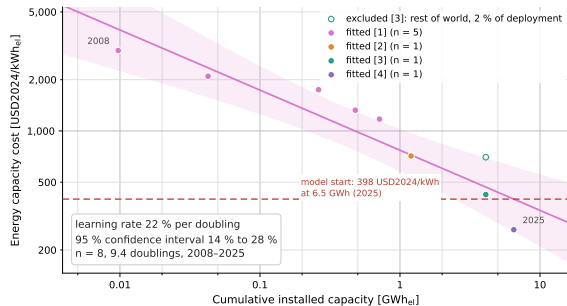
Technology cost and learning

PyPSA Labs, Google

28 September 2026



Vanadium flow battery (multi-day storage)

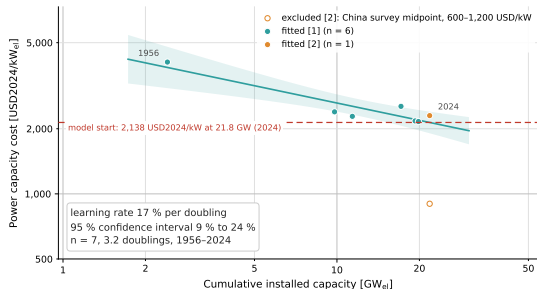
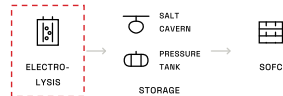


[1] Schmidt et al. (2017), Nat. Energy, 2018 update [2] Dalian Rongke 100 MW / 400 MWh, pv magazine (2022) [3] BNEF (2024) installed flow battery cost via Energy-Storage.news [4] Rongke Xinjiang 200 MW / 1 GWh unit price, Project Blue (2025)

- **Learning.** Wright's law: least-squares fit of log cost on log cumulative GWh, slope b ; learning rate = $1 - 2^{-b}$, 95 % confidence interval from the t -distribution. Cross-check: the Lafond / Way estimator on year-to-year differences (17 %, step noise $\hat{\sigma}_\eta = 0.26$).
- **Model starting cost assumption.** We assume 193 USD₂₀₂₄/kW (technology-data v0.15, from PNNL 2022) next to 398 USD/kWh for the energy part, coinciding with the top-down estimate from the plot. The same learning rate is applied to both parts, consistent with Schmidt et al. (2017).

Long-Duration Storage: charging

Alkaline electrolyser

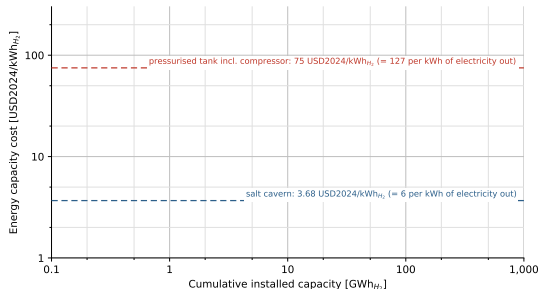
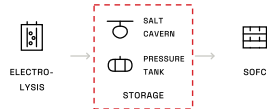


[1] Schmidt et al. (2017), Nat. Energy, 2018 update [2] IEA (2025), Global Hydrogen Review

- **Learning.** The only long cost record for electrolyzers is the alkaline series in Schmidt et al. (2017): six plant prices between 1956 and 2014. We add the IEA Global Hydrogen Review 2024 cost outside China (2,300 USD/kW) at the series end plus the 2 GW operating by end-2024, as the IEA counts the fleet, not cumulative production. Seven points over three doublings give 17 % per doubling, 95 % confidence interval 9 to 24 %. The China midpoint (900 USD/kW) is not fitted: one survey number would set the rate on its own.
- **Model starting cost assumption.** We start the electrolyser at 2,138 USD₂₀₂₄/kW, the fitted line at the last point, 21.8 GW. The catalogue value of 2,492 USD/kW (technology - data v0.15) is 17 % higher but inside the fit's uncertainty. Efficiency 58.7 % on the lower heating value (Danish Energy Agency, alkaline 100 MW); the learning rate applies to this cost.

Long-Duration Storage: storage

Pressurised tank or salt cavern



[1] Danish Energy Agency, Technology Data for Energy Storage, sheet 151a (tanks incl. compressor, 68 EUR₂₀₂₅/kWh, 28 a) [2] same catalogue, sheet 151c (caverns, 3.3 EUR₂₀₂₅/kWh, 100 a) [3] via PyPSA/technology-data v0.15.0

- **Learning.** Neither pressurised tanks nor salt caverns have a record of cost against deployment, so no learning is applied to the storage step and its cost stays where the catalogue puts it.
- **Model starting cost assumption.** Hydrogen can be kept in pressurised steel tanks or in salt caverns. Tanks with their compressor cost 68 EUR₂₀₂₅ per kWh of hydrogen (Danish Energy Agency storage catalogue, sheet 151a, via technology-data v0.15), which is 127 USD₂₀₂₄ per kWh of electricity that later comes out of the fuel cell. Salt caverns cost 3.3 EUR per kWh of hydrogen (sheet 151c) and last a century, so they are about twenty times cheaper, but they need salt formations, which exist in northern Europe, on the US Gulf Coast and in a few other basins. Tanks are the option every archetype can build; caverns are the option for the archetypes that have the geology.

Long - Duration Storage: storage

Global Distribution of Salt Cavern Availability

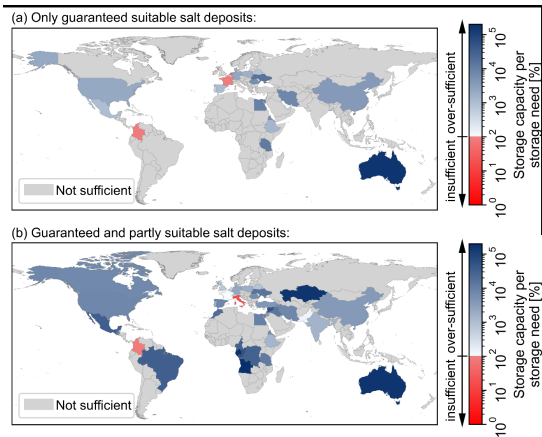
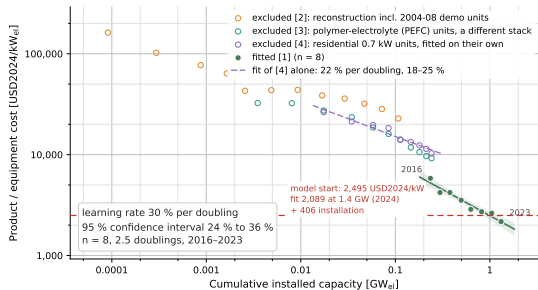


Figure 3 from Franzmann, Schubert, Heinrichs, Kukla & Stolten (2025), Energy Storage Autonomy in Renewable Energy Systems Through Hydrogen Salt Caverns, arXiv:2504.12135

Long-Duration Storage: discharging

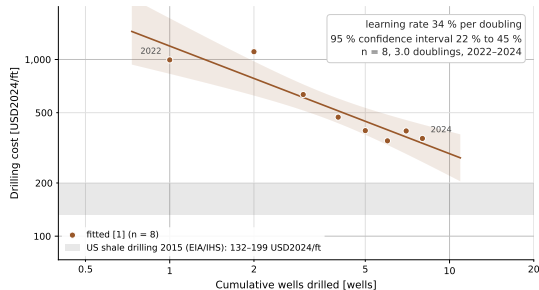
Solid oxide fuel cell



[1] Bloom Energy 10-K key operating metrics (2016-2023)
 [2] Schmidt et al. (2017), Nat. Energy, 2018 update [3] ACE ENE-FARM shipment statistics (2025), PEFC units [4] ACE ENE-FARM shipment statistics (2025), SOFC units, with unit prices from Japan's ANRE

- **Learning.** Bloom Energy's [SEC filings](#) report the manufacturing cost per kW of the fuel cells delivered each year, 2016 to 2023, as deployment grew from 0.24 to 1.3 GW: a learning rate of 30 % per doubling (24 to 36 %). The fleet runs on natural gas; Bloom has run the same units on pure hydrogen in a [100 kW demonstration](#). The dashed line is Japan's residential ENE-FARM programme ([ACE statistics](#)) on its own: 22 % (18 to 25 %), a cross-check from a different product.
- **Model starting cost assumption.** We start at 2,495 USD₂₀₂₄/kW: the fitted manufacturing cost read at the 1.4 GW deployed that Bloom reports in its [10-K for 2023](#), plus the 394 USD/kW installation cost from the same filing. Bloom's margin is not included; its 2020 price was about 4,100 USD/kW. On hydrogen the unit converts at 59 % efficiency ([Bloom, 2024](#)).

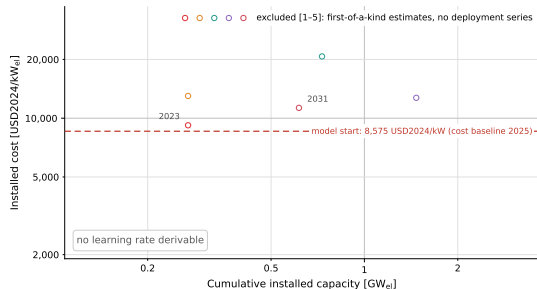
Enhanced geothermal: drilling cost



[1] El-Sadi et al. (2024) via Akindipe et al. (2025)

- **Learning.** Fervo's eight horizontal wells 2022–2024: drilling cost per foot against cumulative wells (Akindipe et al. 2025): 34 % per doubling (22–45 %), the 35 % Fervo reports. The gain sits in wells 2 to 4; the last three are flat at 350–390 USD₂₀₂₄/ft, twice the US shale drilling cost of 2015 (grey band, EIA).
- **Model starting cost assumption.** Plant 14,479 USD₂₀₂₄/kW (NREL ATB 2024, deep EGS binary, moderate); Fervo's first-of-a-kind about 7,000 USD/kW. Drilling is half of plant cost, so the plant-level rate is still to be set.
- **Caveat.** Way et al. (2022) find no long-run decline in geothermal plant cost; Fervo's gain is one operator catching up with shale drilling practice, so the 34 % suits a breakthrough scenario rather than the reference case.

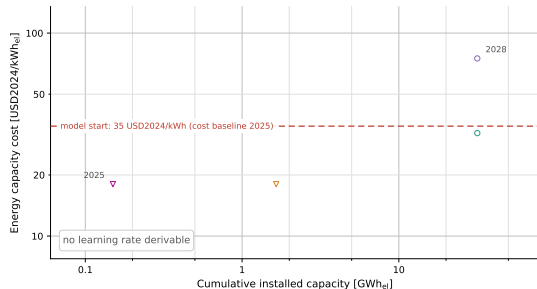
Small modular reactor



[1] EIA Annual Energy Outlook 2025 [2] DOE Liftoff: Advanced Nuclear (2024) [3] NuScale Utah project estimate (2023) [4] Ontario Power Generation, Darlington BWRX-300 [5] TerraPower Natrium Kemmerer 1

- **Learning.** Not enough data to measure learning. Five first-of-a-kind estimates at 9,000–21,000 USD₂₀₂₄/kW: [EIA Annual Energy Outlook 2025](#), [DOE Liftoff Advanced Nuclear 2024](#), NuScale's Utah project (cancelled 2023), Ontario Power Generation's Darlington BWRX-300, TerraPower Natrium; placed at an indicative cumulative capacity (operating fleet plus the project).
- **Model starting cost assumption.** 8,575 USD₂₀₂₄/kW (NREL ATB 2024, small nuclear 300 MWe, moderate case), between first- and nth-of-a-kind. Assumed learning rate 10 % per doubling as a scenario value: large reactors historically show negative to 6 % ([Rubin et al. 2015](#)).

Iron - air battery

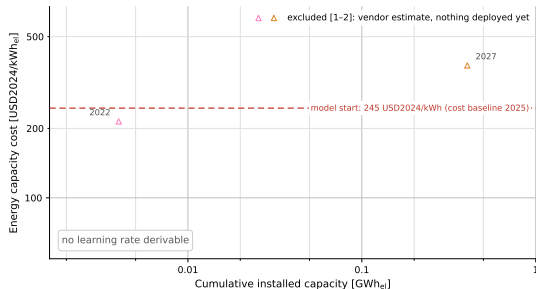


- [1] Form Energy (2023), modelling recommendations [2] Form Energy (2023), vendor target [3] The Information via TechCrunch (2026) [4] Weaver, ESS News (2026)

- **Learning.** Not enough data to measure learning. The two low points are [Form Energy's](#) all-in installed cost target, 15–20 USD/kWh at 100 h for gigawatt manufacturing scale around 2030, placed at the first projects (Great River Energy 150 MWh, 2025; Georgia Power 1,500 MWh, 2026). The first disclosed price is Google's 300 MW / 30 GWh Minnesota system, about 1 bn USD ([The Information via TechCrunch, 2026](#); unconfirmed): 33 USD/kWh after US tax credits, about 77 before them ([Weaver, ESS News 2026](#)); delivery 2028–31.

- **Model starting cost assumption.** 35 USD₂₀₂₄/kWh ([technology - data v0.15](#)): Form's 2030 target back-cast to 2025 at 1.5 times, as Form says first projects cost more; the Google price brackets it. Efficiencies from the same Form table: 70 % charge, 59 % discharge, 41 % round trip. Assumed learning rate 15 % per doubling, inside [DOE LDES Liftoff \(2023\)](#)'s 12–18 %.

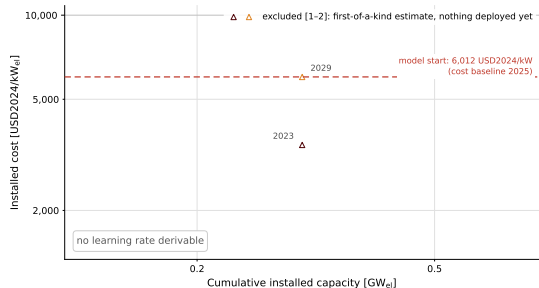
CO₂ battery



[1] Energy Dome, CEO estimate [2] Alliant Columbia Energy Storage (Energy Dome)

- **Learning.** Not enough data to measure learning. Energy Dome's price for first full-scale plants, 220 EUR/kWh installed with EPC and including its margin (2023, [Energy-Storage.news](#)), placed at the Ottana pilot (2.5 MW / 4 MWh, 2022), and Alliant's Columbia Energy Storage project (20 MW / 200 MWh, 300–450 USD/kWh, 2027).
- **Model starting cost assumption.** 245 USD₂₀₂₄/kWh at 10 h, the Energy Dome figure above. Round-trip efficiency 75 % AC to AC, the figure Energy Dome quotes and Alliant's 73–75 % ([Energy Dome](#)); no charge/discharge split is published. Assumed learning rate 5 % per doubling, as for mechanical storage: [Schmidt et al. \(2017\)](#)'s pumped-hydro series gives $-1 \pm 8 \%$.

Allam - cycle gas with carbon capture



[1] NET Power, Project Permian 2023 estimate [2] NET Power, Project Permian SN1 update

- **Learning.** Not enough data to measure learning. NET Power's Project Permian, the first utility-scale plant (300 MW): 3,431 USD₂₀₂₄/kW in the 2023 estimate and 6,009 USD/kW in the 2025 update (USD 1.7–2.0 bn, start 2029; [NET Power Q4 2024 results](#)). The La Porte demonstration (2018) has no disclosed cost.
- **Model starting cost assumption.** 6,012 USD₂₀₂₄/kW, the updated first-of-a-kind estimate. Efficiency 59 % net on the lower heating value with near-complete capture, the design figure ([Allam et al. 2017](#)); no measured value has been published. Assumed learning rate 5 % per doubling, following 2–7 % for CO₂ capture units ([Rubin et al. 2015](#)); fuel sets the cost floor, so capex learning matters less than for storage.